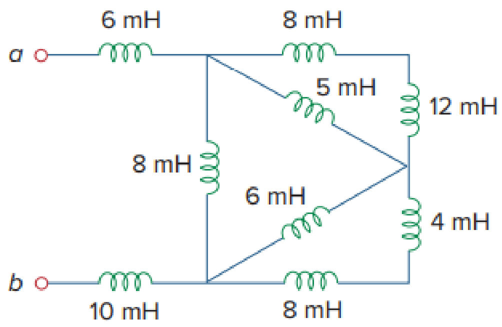
Problem

Find *Leq* at the terminals of the circuit in Fig. 6.75.

Fig. 6.75:



Step-by-step solution

Step 1 of 5

Refer to the circuit diagram depicted in Figure 6.75 in the textbook.

It is clear that, in Figure 6.75, at extreme right end inductors 8 mH and 12 mH are connected in series. Similarly, inductors 8 mH and 4 mH are connected in series.

The equivalent inductances are,

$\Rightarrow 8\text{ mH}+12\text{ mH}$

$\Rightarrow 20\text{ mH}$

And

$\Rightarrow 8\text{ mH}+4\text{ mH}$

$\Rightarrow 12\text{ mH}$

Draw the modified circuit diagram.

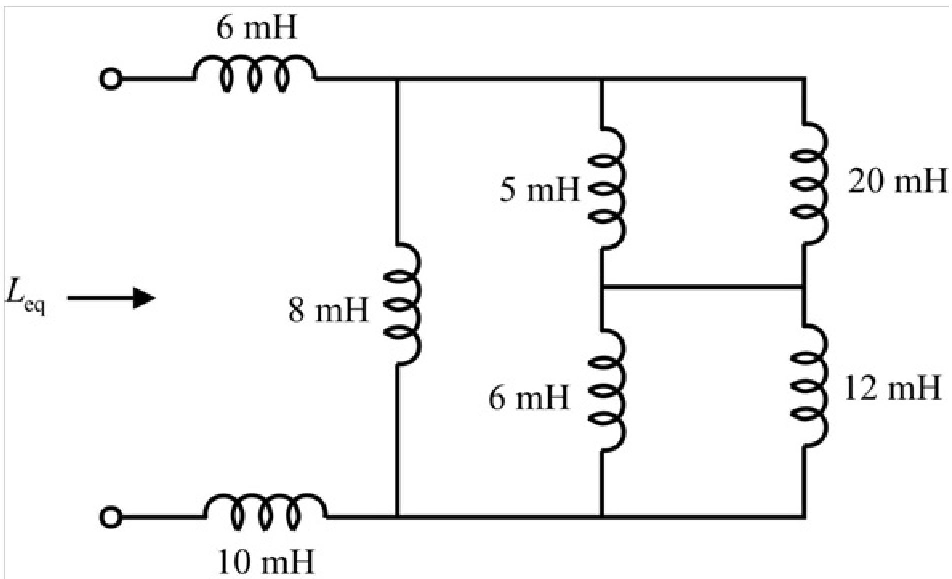


Figure 1

Step 2 of 5

In Figure 1, the inductors $5\ \text{mH}$ and $20\ \text{mH}$ are connected in parallel. Find its equivalent inductance.

$$\Rightarrow 5\,||\,20$$
$$\Rightarrow \frac{5\times 20}{5+20}$$
$$\Rightarrow 4\ \text{mH}$$

Similarly, the inductors $6\ \text{mH}$ and $12\ \text{mH}$ are connected in parallel. Find its equivalent inductance.

$$\Rightarrow 6\,||\,12$$
$$\Rightarrow \frac{6\times 12}{6+12}$$
$$\Rightarrow 4\ \text{mH}$$

Step 3 of 5

Draw the reduced circuit diagram.

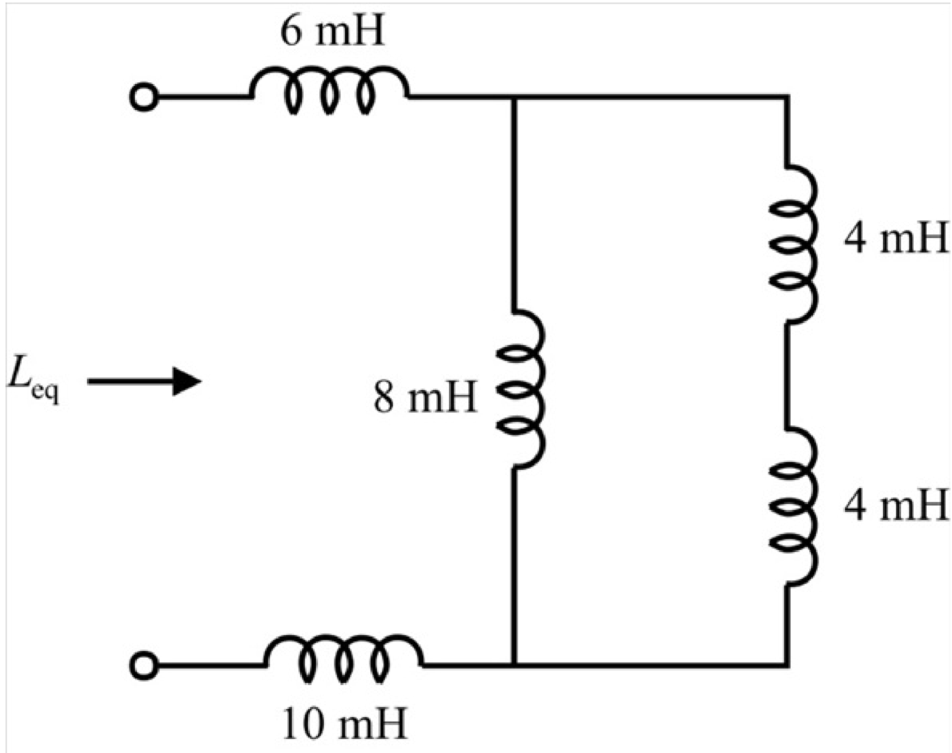


Figure 2

Step 4 of 5

In Figure 2, the series combination of two $4\ \text{mH}$ inductors is in parallel with $8\ \text{mH}$ inductor. Determine its equivalent inductance.

$$\Rightarrow 8 \| (4+4)$$
$$\Rightarrow 8 \| 8$$
$$\Rightarrow \frac{8\times 8}{8+8}$$
$$\Rightarrow 4\ \text{mH}$$

Draw the modified circuit diagram.

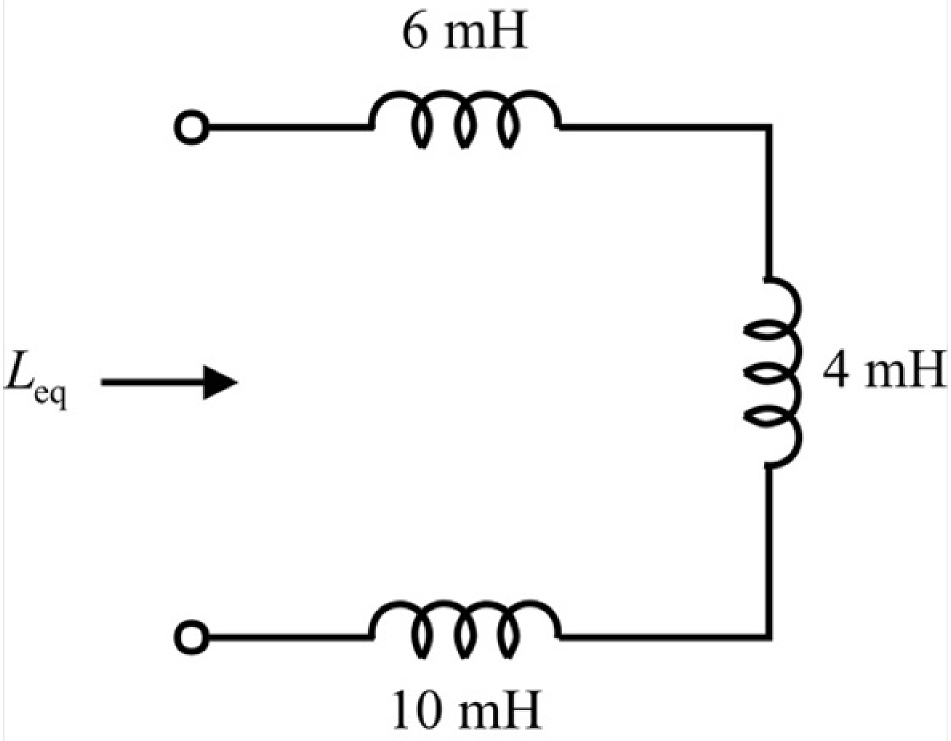


Figure 3

Step 5 of 5

In Figure 3, the three inductors are connected in series. Calculate the equivalent inductance.

$$\begin{aligned} L_{eq} &= 6+4+10 \\ &= 20\ \text{mH} \end{aligned}$$

Therefore, the equivalent inductance of the circuit is $\boxed{20\ \text{mH}}$.